Group 1 Project Final Report

1. Introduction

Machine learning-based algorithms have been widely deployed in the healthcare industry to predict factors like healthcare costs and health insurance claims. While the use of these algorithms often promises efficiency, fairness, and high accuracy, optimized predictive technologies have led to social harms that exacerbated the conditions of marginalized demographic groups (Wang et al., 2023). For example, a study in 2019 has shown how racial bias in an algorithm used to predict potential healthcare needs for patients led to fewer resources being allocated towards African American patients compared to White patients (Obermeyer et al, 2019). Additionally, the same study found that such algorithms often fail to account for historical bias and the decreased opportunities/resources allocated to these marginalized populations, leading to oversights regarding their level of risk and under-assessments in the urgency of their need for care.

Inspired by these phenomena, our project aims to identify how biases during data collection affect algorithmic bias. We will focus our examination using three different model flexibilities: Linear Regression with low flexibility, Polynomial Regression with medium flexibility, and K Nearest Neighbors Regression with high flexibility. Using these algorithms, we will answer the research questions: **How do different biases (representation and historical) during data collection and the flexibility of a model affect algorithmic biases when predicting a patient's hospital billing amount?** From this research, we expect to find that bias during data collection negatively impacts the predictive accuracy of billing amount, specifically for patients within marginalized groups.

2. Data

Our data was collected from a Kaggle dataset titled 'Healthcare Dataset' created by Prasad Patil and contains 10,000 synthetic patient entries (i.e, randomly generated values). Synthetic refers to data that is inorganically generated as opposed to being collected from real individuals or patients in this context. The data set contained variables of interest such as Blood Type, Medical Condition, Insurance provider, Medication, Test Results, and Admission type, and our outcome of interest: patient billing amount.

In terms of data cleaning and preparation, given the synthetic nature of our data, we were able to skip past many of the typical steps needed for data cleaning. For instance, we did not need to handle N/A, miscellaneous, and/or repeated values/rows. While this simplified the data cleaning process, we found other cleaning tasks that needed to be completed. For example, converting all patient names in the 'Names' column to title case, and removing columns 'Room Number' and 'Doctor' that we deemed unnecessary for our analysis. The next step was creating features we hypothesized were correlated with a patient's billing amount. This included the addition of a ‘Race’ column, which we derived for each patient using US census data regarding the demographic makeup of the United States (e.g, 60% White, 12% African-American, etc.). Additionally, a patient’s Admission length (Date of Admission - Date of Discharge) seemed likely to be correlated with their billing amount, so we added an ‘Admission length’ column. Finally, we wanted to analyze potential age biases that could occur during data collection, so we categorized each patient via the ‘Age’ column in buckets ranging from 0-21 to 65+ in the ‘Age Group’ column.

After our initial data cleaning and preparation, we decided on our predictors of Gender, Blood Type, Medical Condition, Insurance Provider, Admission Type, Medication, Test Results, Race, Age Group, Admission Length, and Age. Given that the data was synthetic, we wanted to explore how each predictor, excluding variables such as hospital and doctor, could be a statistically significant predictor of patient billing amount.The final step of data cleaning/preparation we did was after our initial Exploratory Data Analysis. We discovered that the values within our predictors needed to be converted to either categorical predictors or required dummy variables. Using transformer functions `OneHotEncoder() and ColumnTransformer()` from the sklearn package to preprocess our data, we converted all categorical predictors to have corresponding dummy variables. Then, using a Pipeline, we generated our algorithms using this preprocessed data.

Our initial analysis of models using these predictors revealed that, due to the data being synthetic, there existed no relationship between any predictors and billing amount. Thus, we had to create a correlation. To do this, we researched real-world statistics on how each feature may impact billing amount and incorporated that correlation into our data. For instance, research conducted by the World Economic Forum revealed that women, on average, paid 1.18x more on hospital billing compared to men (Edmond,1), so we multiplied each female’s billing amount by 1.18. For Admission Length. (Please refer to the Appendix section for conclusive descriptions of our modifications.)

Beyond injecting correlation, we also aimed to examine the impact historical bias has on algorithmic performance. Historical bias refers to data reflecting historical trends or inequalities that no longer exist in the current context. For this project, we focused on the disparity of billing that exists between African Americans and White patients. Our research found that, given the same level of illness as White individuals, African Americans, on average, spent $1,800 less for treatment. The first reason for this is due to the correlation between race and income: people of color are statistically more likely to have lower incomes due to the vast discrimination and historical bias still prevalent in our country, and as such are less likely to be able to afford the treatment they need. Secondly, lower-income patients are also less likely to access medical services often due to time restraints from their jobs or the areas in which they live. By incorporating this disparity into our data, we are able to examine how well our algorithms handle this historical bias and how it affects the accuracy of their predictions.

3. Methods

There exists a vast array of approaches that could be taken to analyze our dataset, and we decided to utilize several different algorithms on many different subsets of our data in order to provide a truly comprehensive analysis of any potential bias that might be represented. Using multiple different models allowed us to compare and contrast the different algorithms, which in turn was beneficial in revealing not just the existence of bias, but also the root cause.

The first algorithm we utilized was a Linear Regression Model. This is a fantastic first choice, as it is an incredibly simple and straightforward algorithm that allows for us to create a baseline from which to compare other models. Linear regression assumes a linear relationship between predictors, and allows you to see bias in model weights directly, providing a clear understanding on how each variable influences the outcome.

The second algorithm we chose was a K-Nearest Neighbors model. This is meant to predict outcomes based on data distribution by grouping together similar data points into clusters. KNN is particularly effective at detecting data-driven bias, and by comparing it to our linear regression model, we can see how much of the model's bias comes from the data itself versus the algorithm's assumptions.

The third and final algorithm we employed was Polynomial Regression. This is a non-linear regression model that can outperform standard linear regression at predicting more complex relationships, especially when bias is not uniform across populations. Additionally, it allows us to select our own degree for the polynomial, which means we can get the best possible fit for the data. Something important to note when working with polynomial regression is that this model has a tendency to overfit the data, meaning that due to its adaptability, it can perfectly fit the training data, and follow patterns that are actually just random noise. This accentuates the importance of carefully choosing the most appropriate degree.

After selecting our models, we had to format our data for training and then evaluate each model's performance on biased subsets of our data. First, our models were trained on our original dataset, which matched real-world distributions of Age, Race, and Gender statistics. We then created biased subsets of our dataset, which were built upon unequal splits of each of these demographics. These consisted of Gender split datasets (containing 70, 80, and 90% male participants, respectively), Race split datasets (70, 80, and 90% white participants), and Age split datasets (70, 80, and 90% of participants from the 0-40 age group). To create these datasets while ensuring the original data structure was preserved, each subset was constructed by randomly sampling participants from the original dataset until we had groupings that matched our desired ratios exactly. Once all nine subsets had been created, we ran each of the three predictive algorithms on each of the subsets, measuring the R^2 value, the RMSE (Root Mean Squared Error), and the predicted billing amount/standard deviation for each split. (Root Mean Squared Error is a metric that measures the average magnitude of errors between predicted values and actual observations in a model. It indicates how spread out the prediction errors of a model are, with lower values suggesting better accuracy.)

By utilizing this split-dataset approach, we are able to perform controlled experiments on demographic imbalance. When we examine datasets with specific demographic ratios, we can see how the models behave as the underlying population structure changes: For example, if our model predictions change drastically with a 90/10 gender split as compared to a 50/50, we can extrapolate that gender is a sensitive characteristic that has influence over model predictions, and this raises red flags for fairness.

This approach is also useful for separating data-driven bias from algorithm-driven bias. Each of these models handles demographic imbalance differently, and by keeping the training data fixed and only modifying the testing population, we can determine if bias originates from the structure of the algorithm, the structure of the data, or the interaction between the two.

In addition to investigating representation bias, we also took steps to investigate how historical bias would affect our results. As previously discussed in our data section, research found that given the same level of illness as White individuals, African Americans, on average, spent $1,800 less for treatment (Obermeyer et al, 2019). We incorporated this result by creating a dataset in which we calculated the billing amount using the same metrics as before, but simply decremented the billing value of rows in which the "Race" variable was equivalent to "African American" by a value of 1800. We then ran our three different models on both the original dataset and the new decremented dataset to compare the R^2 value, the RMSE value, and, most importantly, the mean billing amount.

4. Results

To answer our research question, we analyzed the effects of three distinct data collection biases (Age, Race, and Gender) on the predictive accuracy and fairness of three models with varying flexibility: **Linear Regression**, **Polynomial Regression** (Degree 2), and **KNN Regression** (k=15). Our analysis focused first on aggregate predictive accuracy (R-Square and MSE/RMSE) and second on algorithmic bias (group-specific error disparities).

4.1 Aggregate Model Performance

As shown in Figure 1, the Polynomial Regression model was consistently the most accurate across all analyses. When trained on the original, unbiased data, it achieved the highest R-Square (≈ 0.99) and the lowest Mean Squared Error (MSE) (≈ 3000 to 8000). The low-flexibility Linear Regression model was also a strong performer (R-Square ≈ 0.90), while the high-flexibility KNN model was the weakest (R-Square ≈ 0.88).

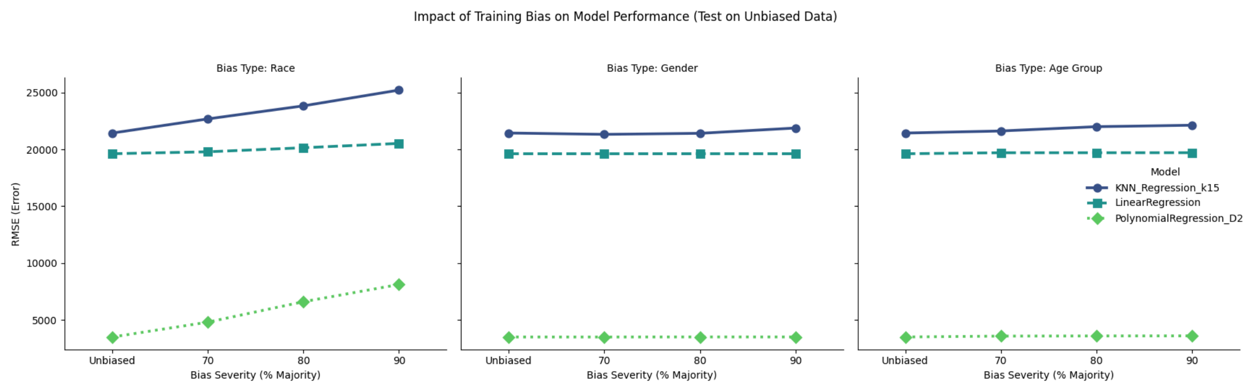


Figure 1: This faceted point plot shows how the Root Mean Square Error (RMSE) changes for three models (Linear, Polynomial, KNN) as the training data becomes increasingly biased (from Original to 90% Majority). Each panel represents a different demographic bias (Race, Gender, Age). Upward trends indicate that the model is losing accuracy as bias increases.

4.2 Impact of biased datasets

We then investigated how introducing bias into training data affected the performance of these models.

- **Age Group Bias**: We trained all three models on datasets with increasing age bias (70/30, 80/20, and 90/10 splits favoring the 0-40 age group). The overall accuracy of all three models remained remarkably stable. The **Polynomial** model's RMSE was ≈ 3500, the **Linear** model's was ≈ 19,000, and the **KNN** model achieved RMSE ≈ 21,000 regardless of whether they were trained on the unbiased data or different biased splits.
- **Gender Bias:** Similarly, for the gender analysis, we trained one set of models on the original data and used them to predict on test sets with 70/30, 80/20, and 90/10 Male/Female splits. The mean predicted billing amount remained almost constant (e.g., ≈ 19,000 for Linear Regression, ≈ 21,000 for KNN, ≈ 3,500 for Polynomial), showing no significant change in the average prediction despite the skewed test data.
- **Racial Bias**: Interestingly, unlike the Age and Gender groups, where severe representational bias did not propagate into significant performance degradation, we observe a different pattern in the results trained on racial bias datasets. As the training data become increasingly skewed toward the majority group (White), the model's ability to generalize to the unbiased data deteriorates noticeably. For example, the polynomial regressions saw their error more than double, jumping from an RMSE of 3,500 on the unbiased set to nearly 8,000 when trained on the 90% biased set. Similarly, the K-Nearest

Neighbors (KNN) model showed a steady decline in performance, with RMSE rising from ≈ 22,000 to over 25,000. Even the Linear Regression model, while less significantly affected, exhibited a consistent upward trend in error. This suggests that for this specific prediction task, racial diversity in the training data is a critical factor for maintaining overall model accuracy, whereas the models were largely resilient to imbalances in age and gender.

4.3 Algorithmic Bias

Finally, we analyzed algorithmic bias by calculating the error (RMSE) for each demographic group when testing on the unbiased dataset. This allowed us to determine if models produced more accurate predictions for some groups than others and if the skewed training data propagated bias into the final predictions.

Age & Gender Bias: Contrary to the idea that a severely skewed dataset would lead to worse performance for underrepresented groups, the age and gender representation we introduced into the training data did not result in propagated bias. As shown in the figure below (fig. 2 & fig. 3), We found that the prediction error (RMSE) for each specific age and gender group remained mostly consistent when trained on datasets with an increasing level of biases.

- Polynomial Regression: RMSE of ≈ 3,500.
- Linear Regression: ≈ 19,000.
- KNN RMSE of ≈ 21,000.

These consistent scores across different training splits indicate that for Gender and Age, the models were resilient to the representational shifts in the training data.

Racial Bias : Similar to the pattern we observed in Section 3.1, the racial bias analysis revealed a significant sensitivity to training data composition. Unlike the stability seen in age and gender, shifting the training data towards the majority group (from Unbiased to 90%) caused noticeable performance degradation across the models when evaluated on the unbiased test set.

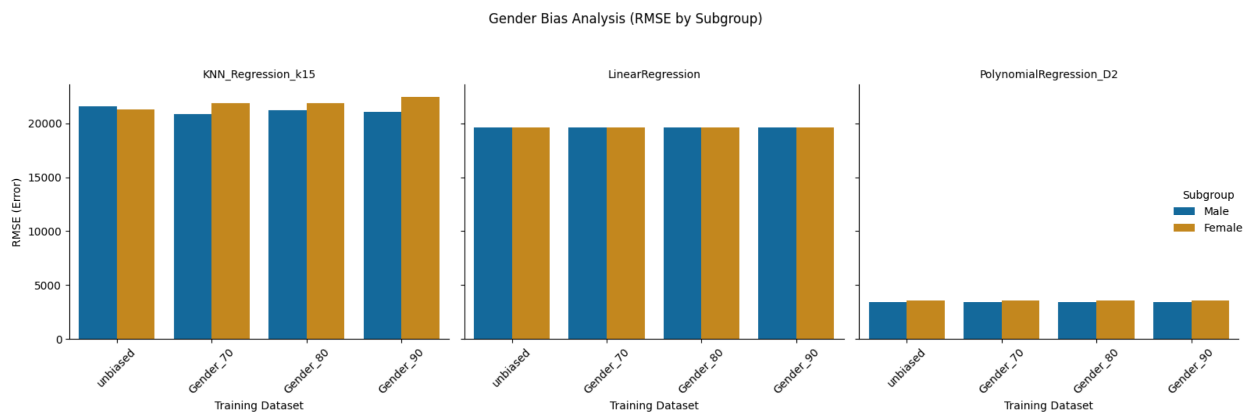
- **Polynomial Regression:** Despite being the most accurate model overall, it exhibited the most dramatic sensitivity to racial skew. The RMSE for White patients more than tripled, jumping from **≈ 2,981** on unbiased data to **≈ 9,876** when trained on the 90% biased dataset. This suggests that the model overfitted to the specific label distribution of the biased training set, failing to generalize even for the majority group it saw most often.
- **K-Nearest Neighbors (KNN)**: This model showed a broad decline in performance for both majority and minority groups. As the training data became less diverse, the error for White patients increased from ≈ 21,310 to ≈ 24,766,

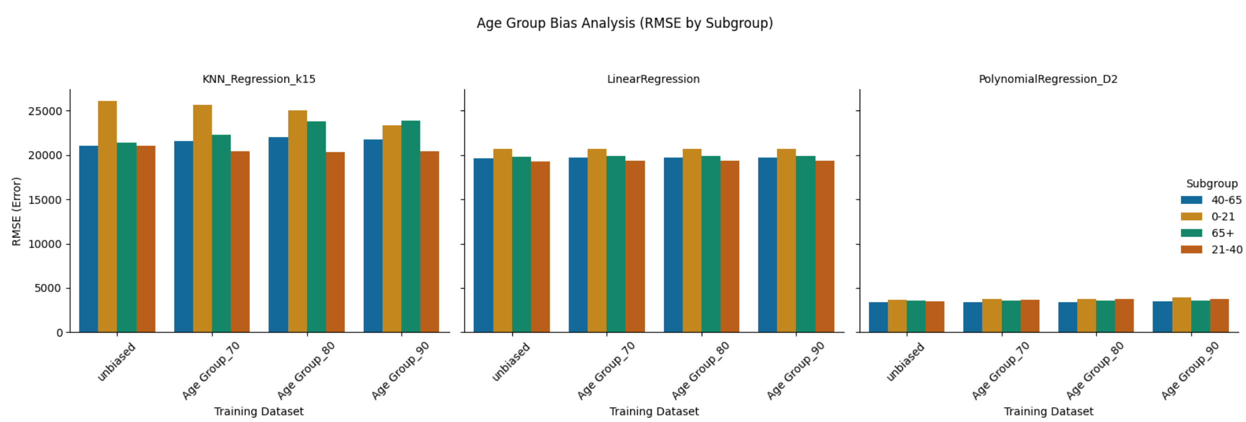
while error rates for underrepresented groups like Asian patients spiked from ≈ 28,129 to ≈ 37,720.

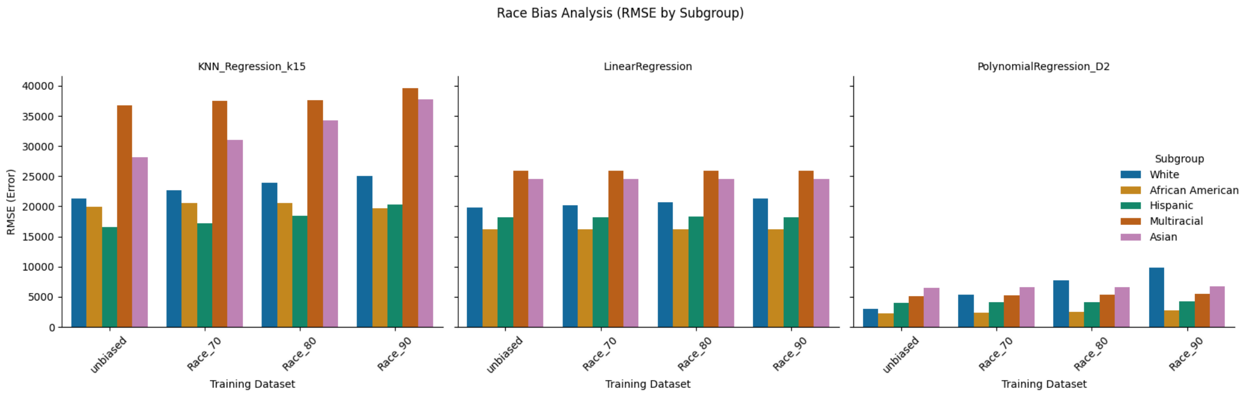
- **Linear Regression:** This model remained the most robust, with error rates for White patients increasing only slightly from **≈ 19,829** to **≈ 21,302**.

Our analysis also reveals a critical trade-off between model accuracy and fairness. While Polynomial Regression achieved the highest overall accuracy with an RMSE of approximately 3,500 compared to 19,000 for Linear Regression, it demonstrated significant inherent bias even on unbiased data, predicting costs for White and African Americans far more accurately than for underrepresented groups like Asians and Multi-Racial. Conversely, KNN emerged as the least effective model, suffering from instability, high sensitivity to data bias, and substantial fairness gaps between racial groups.

Figures 2 to 4: Group-specific RMSE on the unbiased test set for models trained on datasets with varying racial bias. Note that for each model (e.g., PolynomialRegression_D2), the error for each group (e.g., 'White') is identical across all four trained models, from 'unbiased' to 'biased_90'.







4.4 Historical Bias

Finally, we examined the RMSE and average predicted billing amount of our historical bias dataset using a linear, KNN, and polynomial regression model. From our algorithms, we saw that the RMSE was about $200 higher for unbiased data than for biased data for Linear and KNN Regression, and about $900 higher for polynomial regression. The average RMSE for unbiased data was about $14,602 compared to about $14,159 for biased data. Additionally, the average predicted billing amount for all algorithms was roughly $4,300 greater for the unbiased data compared to the biased historical data, with the mean average prediction being $104,704 for the unbiased data and $100,397 for the historical data.

5. Discussion:

5.1 Findings

Our analysis aims to answer how data collection biases (Race, Gender, and Age), historical bias, and model flexibility affect algorithmic bias in healthcare billing amount predictions. Our analysis led to three primary conclusions.

5.2 Data Collection Bias

Our first major finding revealed distinct constraints on how different types of representational bias affect model performance. For Age and Gender, data collection bias had a negligible effect on both aggregated accuracy and group-specific fairness. Contrary to the intuition that skewed training data would automatically lead to degraded performance for underrepresented groups, we did not observe propagated bias in these categories. The prediction error (RMSE) for each age and gender group remained remarkably stable, regardless of whether the models were trained on unbiased data or datasets with extreme 90/10 skews

However, Racial Bias presented a different challenge. Unlike Age and Gender bias, data collection bias resulted in noticeable performance degradation when it comes to the model's ability to generalize to the unbiased dataset. As the training data became increasingly skewed towards the majority group, the predictive error increased significantly. This finding confirms that for our specific task, the model relies on Race/Ethnicity as the primary factor for decision making, demonstrating how racial diversity is critical for maintaining model reliability.

5.3 Model Flexibility

Second, we found that model flexibility was the primary driver of both overall accuracy and algorithmic fairness, often in conflicting ways. While propagated bias was only observed in the racial analysis, inherent bias is evident across all three models. Specifically, for highly flexible models like Polynomial Regressions demonstrated the complexity of this trade off. While Polynomials achieved the highest overall accuracy, they also exhibited significant inherent disparities and extreme sensitivity to racial skew where the error rates for both the majority and underrepresented minority groups spiked exponentially when trained on biased data. In contrast, while less flexible models might be less accurate, they successfully maintain consistent error rates across different demographic groups and remain robust to the severe representational skews. These findings suggest how optimizing for aggregate performance metrics can sometimes obscure significant inherent biases in a model's performance. A model that performs best in overall accuracy may not be the most appropriate choice when fairness and equitable outcomes are our primary concerns.

5.4 Historical Bias

We hypothesized that the average prediction using the historical biased dataset would be lower than the average prediction from the unbiased dataset. From the results, we can see that our hypothesis was confirmed as the average prediction on the unbiased data was about $4,300 more compared to the biased data. Our project focused on addressing the historical notion that African American patients are less likely to seek out medical care due to a lack of resources. From the results, we showed that the inclusion of this historical bias led to lower predicted billing amounts for African American patients. This increases the risk of African Americans being neglected in receiving proper care for illnesses.

5.5 Connection to Literature

The Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification (Buolamwini & Gebru, 2018) study demonstrated that facial recognition algorithms exhibit catastrophically higher error rates on dark-skinned women due to representational disparities in training data. While our study initially sought to determine whether undersampling would yield similar results for minority groups in medical billing amount prediction, our results revealed a more nuanced reality.

Contrary to the Gender Shades findings, where under-representation hurts accuracy, we found that under-sampling does not automatically lead to higher error rates for minority groups. Instead, the harm from undersampling largely depends on the model architecture's flexibility. This suggests that while representational disparity is a root cause of bias, its capacity to harm predictive performance is often tied to model choice.

5.6 Limitations and Future Work

While this analysis was conducted to the best of our ability, there is always room for improvement and areas that require further evaluation. The greatest limitation of all was the fact that the data we worked with was synthetically generated. We took many steps to ensure that our dataset was still viable for exploring bias, such as creating demographic distributions based on real-world data, and assigning weights to our categorical variables to create correlation based on research as to how they should affect our outcome variable, but there is no way to perfectly recreate the precision of actual data, and this must be kept in mind when reviewing our results.

Additionally, while the models we utilized had many advantages, they also have their own limitations and weaknesses. Linear regression is great because it is simple, but it also has a tendency to oversimplify complex relationships and assume additivity/linearity. The KNN model is very sensitive to data imbalance and has the potential to overfit majority clusters or distort minority predictions. Polynomial regression is adept at handling complex relationships, but risks overfitting, and has a high sensitivity to outliers. If we were to return to this project in the future, it would be beneficial to compare more models in addition to these three, as a way to offset some of these limitations. For example, we could incorporate random forests, as they have very low overfitting, as well as neural networks, due to their extensive ability to detect all possible interactions between variables.

One big issue with our methods relates to the way in which we split the data. We selected a 70/80/90% split to examine how our models would perform on extremely biased datasets, and while this was beneficial, we could have learned even more by incorporating a wider range of comparisons, both for extremes such as 99/1% splits or 100/0%, but also for reversing our splits. For example, we could observe the difference in model performance for 80% female participants vs. 80% male participants.

While our approach successfully demonstrates how model predictions can shift under changes in demographic composition, by incorporating a wider range of models, as well as a greater variation in the way we split our biased data, we would be able to significantly strengthen our findings and provide even more definite results.

5.7 Ethical and Societal Implications

Given our dataset was synthetically generated, there were no ethical violations in the data collection process itself, such as concerns regarding the data collection process violating patient privacy. However, because we introduced correlation into the data, we have potentially introduced the risk of negative societal implications. For instance, to introduce correlation, we conducted surface-level research (i.e, browsing the internet) to find trends between our features and outcome. Without conducting proper experiments or citing published research regarding the correlation between features and outcomes, we are potentially misrepresenting how different traits correlate with billing amount. Therefore, our work may suggest certain groups are being wrongfully discriminated against by the healthcare industry, in turn making them less willing to seek out proper care.

However, because we are able to manipulate our data to discover correlations, it allows us to also see the impact different biases have on predictive accuracy. We are able to replicate potential real-world biases without needing to access real patients' information and critique the harms that come with different biases. Additionally, because we are the ones manipulating the data, the issue of accountability is designated to us or any future users. We know which correlations or biases are injected into the data; therefore, the algorithm's results and any effects, good or bad, are also communicated to us. Our work serves as a tool for validating hypotheses and a warning about the harms of using biased data to train algorithms.

5.8 Project Reflection

Given the opportunity to approach this project from the beginning, there are several key aspects that we would have reconsidered. Most notably, while it was an interesting and enlightening experience to work with synthetic data, we would spend time digging deeper into ways we could work with ethically collected patient data without violating personal privacy. We also spent the vast majority of our time focusing on different types of representation bias in our data and algorithms; it would have resulted in a more well-rounded project if we had devoted more of our time to analyzing various types of biases (such as historical or sample bias). Looking back on our results and the poor performance of the KNN model, we would have benefited from exploring additional types of algorithms, such as neural networks or regression trees. Overall, given the data and methods we selected, we are quite satisfied with both the results and implications of our project. The skills and insights gained from this project will certainly provide great value in our future endeavours.

6. Appendix:

Correlation between predictors and outcome: Billing Amount			
Predictor	**Correlation w/ Outcome**	**Adjustment in Data**	**Source(s)**
Gender	Women's Billing Amounts on average are 18% higher than that of Men.	If patient gender = Female, Billing Amount * 1.18	https://www.weforum.org/stories/2023/10/healthcare-equality-united-states-gender-gap/
Admission Length	Average cost per day of hospital visit is $3,025	Find total cost of stay using Admission Length x $3,025 and Add to current Billing Amount	https://nchstats.com/average-cost-of-hospital-stays-in-us/
Medical Condition	**Diabetes:** Roughly $19,736 annual average medical expenditures(2022) translates to $54/day	Admission Length x 54 Added to the current patient Billing Amount	https://pubmed.ncbi.nlm.nih.gov/37909353/
	Asthma: Roughly $3,266 annual per-person medical cost (2015 dollars) translates to $9/day	Admission Length x 9 Added to the current patient Billing Amount	https://cms.illinois.gov/benefits/stateemployee/bewell/financialwellness/economic-impact-of-asthma.html?
	Hypertension: People with high blood pressure had $2,759 higher annual medical costs, about $7.56/day higher than w/out hypertension	Admission Length x 7.56 Added to the current patient Billing Amount	https://cdc.gov/nccdphp/priorities/high-blood-pressure.html?
	Obesity: Adults with obesity pay on average roughly $2,506 more than normal weight translates to $6.87 more /day	Admission Length x 6.87 Added to the current patient Billing Amount	https://pmc.ncbi.nlm.nih.gov/articles/PMC10394178/?

	Arthritis: Average annual individual cost estimates globally range widely: Roughly $700 to $15,600 per person translates to about $5,000 more per patient or $14/day **$5000 is a good estimated median value	Admission Length x 14 Added to the current patient Billing Amount	https://pmc.ncbi.nlm.nih.gov/articles/PMC8605034/?
	Cancer: Mean cost per stay roughly $22,100(2017) translates to $3,400/day	Admission Length x 3400 Added to the current patient Billing Amount	https://hcup-us.ahrq.gov/reports/statbriefs/sb270-Cancer-Hospitalizations-Adults-2017.jsp
Admission Type	Emergency Room visits are about $2,000 more expensive per visit compared to non-emergency visits	If patient Admission type = Emergency, add 2,000 to Billing Amount	https://www.fastaiduc.com/cost-urgent-care-visit-vs-er?
Race	**Asian:** Baseline $0 additional costs	No change to Billing Amount	https://pmc.ncbi.nlm.nih.gov/articles/PMC8371574/#:~:text=8%2C9%2C10,11 https://www.theguardian.com/society/2019/oct/25/healthcare-algorithm-racial-biases-optum
	Hispanic: About $1,333 more compared to baseline	Multiple Billing Amount by 1.28x for patients with Race = Hispanic	
	Black: $2,669 more compared to baseline. -$1800 Due to historical biases	Multiple Billing Amount by 1.57x for patients with Race = Black Subtract 1800 from final value for patients with Race = Black	
	White: $3,449 more compared to baseline	Multiple Billing Amount by 1.74x for patients with Race = White	
	Multi-Race: $4,584 more compared to baseline	Multiple Billing Amount by 1.98x for patients with Race =	

		Multi-Race	
Age Group	**Under 21:** Ages 21 and under on average spend about $2,000 on healthcare	Add $2,000 to patients billing amount where Age Group = 0-21	https://www.registerednursing.org/articles/healthcare-costs-by-age/
	Age 22-40: Ages 22-40 on average spend about $3,000 on healthcare	Add $3,000 to patients billing amount where Age Group = 22-40	
	Age 45-65: Ages 45-65 on average spend about $6,500 on healthcare	Add $6,500 to patients billing amount where Age Group = 45-65	
	Age 65+: Ages 65+ on average spend about $11,000 on healthcare	Add $11,000 to patients billing amount where Age Group = 65+	

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